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Professor Richard Bucknall
Department of Mechanical Engineering
University College London
Marshgate Building, UCL East
Queen Elizabeth Olympic Park, London E20 3BS, United Kingdom

Dear Professor Bucknall and Members of the Selection Committee,

Application for Lecturer in Control Engineering for Electrified Propulsion Systems, B04-07381

I am writing to apply for the **Lecturer (Grade 8)** position in **Control Engineering for Electrified Propulsion Systems** in the **Department of Mechanical Engineering** at UCL. I am currently a Research Assistant at the **Power Electronics and Machines Centre (PEMC), University of Nottingham**, having completed my **PhD in Electrical Engineering at Southeast University (SEU), Nanjing**, in September 2025 with the dissertation *Multi-physics Analysis and Design of Hairpin Winding PM Traction Machines in Electric Vehicles*. My doctoral and postdoctoral research on **drive-system dynamics, multi-physics modelling, and AI-driven design optimisation of electrified propulsion systems** is well aligned with the post and the mission of the **Advanced Propulsion Laboratory (APL)** at UCL East.

My research programme addresses the dynamic behaviour and control-oriented modelling of electrified drive systems. I have developed **spatial-harmonic inductance models** and **converter-machine interaction analyses** that provide the plant representations needed for model-based and model-predictive control of electric drives, alongside **high-fidelity lumped-parameter thermal networks** validated against AVL dynamometer measurements. In parallel, I have developed an **AI-driven surrogate modelling** capability, in which an implicit-physics deep neural network maps machine design parameters to the full magnetic vector potential distribution ($R^2 = 0.9996$ for torque, 94%+ inference speedup over finite-element analysis), built on an **in-house Python finite-element code package** I developed for electromagnetic, structural-modal and mechanical-stress analysis.

I have also gained substantive **industrial development experience** through a multi-year collaboration with **Inovance**, a world-leading EV traction-machine supplier, delivering two oil-cooled PM traction machines (18 000 rpm, 113 kW; 20 000 rpm, 270 kW) through the full industrial cycle of multi-physics design, engineering-drawing release and AVL dynamometer characterisation. My current PEMC role on the **AIMD programme** (Nottingham, Bristol, Newcastle) covers integrated motor-drive technologies. This combined academic and industrial background supports competitive applications to **EPSRC**, **Innovate UK**, **ATI** and industrial partners.

My research record includes **22 journal papers (17 in IEEE Transactions)**, 18 conference papers and 9 patents, recognised by the **2025 IEEE TEC EV Design Competition 1st Prize**, the **2025 IEEE Trans. Transportation Electrification Prize Paper Award**, the **IEEE IAS CMD Thesis Contest 1st Prize** and the **CIEEC Best Paper Award**. I serve as Co-Chair of an ICEM 2026 Special Session on *Next-Generation Electrified Transport: Advances in Electric Drive System*. I would be glad to contribute to teaching in **control engineering**, **electric machines and drives**, **power electronics** and **propulsion systems** at both undergraduate and postgraduate level, and to support the Department's curriculum in the areas of modelling and control of electrified drive systems.

My CV, publication list, teaching statement, research strategy and supporting statement are attached. I would be glad to discuss my application further at interview.

Yours sincerely,

Wentao Zhang